

TRAINING-PATH DEPENDENCE IN BOUNDED CONVEX DYNAMICS: EXPOSURE PRODUCTS, CLEAN-TAIL RETURN, AND AN AFFINE RECURSION

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ABSTRACT

This paper records three narrowly scoped ways in which a training-path intervention can be analyzed. A fixed quadratic has an exact matrix-product solution whose first-order log contraction depends on cumulative learning-rate exposure, with a finite-step squared-step remainder. A twice-differentiable, strongly convex clean objective contracts full parameter error after an intervention is removed; a faster directional rate requires an additional invariant-subspace condition. An affine scalar conditional-mean recursion has an exact product law and an exponential upper bound. These statements do not imply corresponding laws for logistic regression, shuffled SGD, CNNs, nonconvex optimization, or real label noise. The current public package makes no empirical or deployment claim.

1 INTRODUCTION

Training-data attribution and unlearning study relationships between a training set and a trained model, including influence functions (Koh & Liang, 2017), datamodels (Ilyas et al., 2022), TRAK (Park et al., 2023), and retraining-based unlearning (Bourtoule et al., 2021). This paper addresses a more limited mathematical question: when an objective is changed along a training trajectory, which contraction quantities can be stated exactly?

The answer depends sharply on the model. We prove three results, each with its own assumptions. T1 is a fixed-quadratic product identity. T2 is a strong-convex clean-tail contraction result. T3 is an affine scalar conditional-mean identity. The results share product structure but are not interchangeable: neither T1 nor T3 establishes a law for nonlinear SGD, and T2 is not a machine-unlearning guarantee for arbitrary neural networks.

Scope. The frozen public baseline contained historical numerical material without co-located public code, output, configuration, seed, and preregistration artifacts. That material is not part of this submission source. A separate, uncompiled archive points to the immutable checkpoint containing it. A planned two-scenario experiment is a falsification path only, not evidence.

2 FORMAL SETUP

Let A denote a clean objective and let D denote an intervention block. The results below use only the objective active at the relevant step; they do not assume that D is a particular data type, or that an observed loss signal identifies it. Write $\eta_t \geq 0$ for a deterministic step size and $E_{s:T} = \sum_{t=s}^{T-1} \eta_t$ for tail exposure. We reserve the word return for distance to the minimizer of the clean objective, not for accuracy, a representation, or a deployment outcome.

3 THREE BOUNDED RESULTS

3.1 T1: FIXED-QUADRATIC EXPOSURE PRODUCT

Lemma 1 (T1). *Let*

$Q(w) = \frac{1}{2}(w - w^*)^\top H(w - w^*)$, $0 \prec \mu I \preceq H \preceq LI$,
and run $w_{t+1} = w_t - \eta_t \nabla Q(w_t)$ with $0 \leq \eta_t L < 1$. Then

$$w_T - w^* = \prod_{t=0}^{T-1} (I - \eta_t H)(w_0 - w^*). \quad (1)$$

For an eigenvalue λ_j of H , set $P_j = \prod_t (1 - \eta_t \lambda_j)$, $E = \sum_t \eta_t$, and $q_j = \max_t \eta_t \lambda_j < 1$. Then

$$0 \leq -\log P_j - \lambda_j E \leq \frac{\lambda_j^2 \sum_t \eta_t^2}{2(1 - q_j)}. \quad (2)$$

Proof. Since $\nabla Q(w) = H(w - w^*)$, one step gives $w_{t+1} - w^* = (I - \eta_t H)(w_t - w^*)$, and iteration proves equation 1. Diagonalize the symmetric positive-definite H . On an eigenvector with eigenvalue λ_j , the multiplier is P_j . For $x \in [0, q_j]$,

$$0 \leq -\log(1 - x) - x = \sum_{k \geq 2} \frac{x^k}{k} \leq \frac{x^2}{2(1 - q_j)}.$$

Apply this inequality to $x = \eta_t \lambda_j$ and sum over t . This proves equation 2. $\square$

T1 says that equal exposure yields the same first-order log contraction in this fixed quadratic. It does not say that arbitrary finite schedules with equal exposure have identical terminal iterates: the squared-step term quantifies the permitted discrepancy. Replacing a step η by K steps of size η/K preserves exposure, whereas repeating K original steps changes exposure and is not a finite-budget invariance.

3.2 T2: STRONG-CONVEX CLEAN-TAIL RETURN

Theorem 1 (T2). *Let L_A be twice continuously differentiable, μ -strongly convex, and L -smooth on every segment joining a tail iterate to its unique minimizer w_A^* . If $0 \leq \eta_t \leq 1/L$, then after switching to gradient descent on L_A at step s ,*

$$\|w_T - w_A^*\|_2 \leq \prod_{t=s}^{T-1} (1 - \eta_t \mu) \|w_s - w_A^*\|_2 \leq e^{-\mu E_{s:T}} \|w_s - w_A^*\|_2. \quad (3)$$

If, in addition, $e \neq 0$ and $\text{span}(e)$ is invariant for every segment average Hessian in the proof and its eigenvalue on e is at least λ_e , then

$$|e^\top (w_T - w_A^*)| \leq e^{-\lambda_e E_{s:T}} |e^\top (w_s - w_A^*)|. \quad (4)$$

Proof. For each $t \geq s$, the fundamental theorem of calculus gives

$$\nabla L_A(w_t) - \nabla L_A(w_A^*) = \bar{H}_t(w_t - w_A^*), \quad \bar{H}_t = \int_0^1 \nabla^2 L_A(w_A^* + r(w_t - w_A^*)) dr.$$

The assumed Hessian bounds imply $\mu I \preceq \bar{H}_t \preceq LI$. Therefore

$$w_{t+1} - w_A^* = (I - \eta_t \bar{H}_t)(w_t - w_A^*).$$

All eigenvalues of $I - \eta_t \bar{H}_t$ lie in $[0, 1 - \eta_t \mu]$ because $\eta_t \leq 1/L$. Hence its spectral norm is at most $1 - \eta_t \mu$. Multiply these bounds from s through $T - 1$, then use $1 - x \leq e^{-x}$ to obtain equation 3.

For the directional statement, the added invariance premise makes e an eigenvector of every $\bar{H}_t$. Its scalar multiplier is at most $1 - \eta_t \lambda_e$; the same multiplication and exponential bound prove equation 4. A Rayleigh-quotient lower bound alone does not imply this shared invariant-subspace premise. $\square$

Thus T2 is a full-parameter result under explicit strong-convexity conditions. Its directional refinement is conditional and must not be inferred from observations in another model class.

3.3 T3: AFFINE SCALAR CONDITIONAL-MEAN PRODUCT

Theorem 2 (T3). *Let (u_t) be an integrable scalar process adapted to $(\mathcal{F}_t)$. Fix $\lambda > 0$ and deterministic step sizes satisfying $0 < \eta_t \lambda < 1$. Suppose*

$$u_{t+1} = (1 - \eta_t \lambda)u_t + \eta_t \xi_t, \quad \mathbb{E}[\xi_t | \mathcal{F}_t] = 0. \quad (5)$$

Then for any $s < T$,

$$\mathbb{E}[u_T] = \mathbb{E}[u_s] \prod_{t=s}^{T-1} (1 - \eta_t \lambda), \quad |\mathbb{E}[u_T]| \leq |\mathbb{E}[u_s]| e^{-\lambda E_{s:T}}. \quad (6)$$

Proof. Taking conditional expectations in equation 5 and then total expectations yields

$$\mathbb{E}[u_{t+1}] = (1 - \eta_t \lambda) \mathbb{E}[u_t].$$

Repeated substitution gives the product identity in equation 6. Every factor is positive and $1 - x \leq e^{-x}$ for $x \geq 0$, so multiplying the inequalities gives the exponential upper bound. $\square$

T3 is exact for the displayed scalar recursion. It does not assert that logistic-SGD, random reshuffling, or a nonlinear accuracy functional obeys equation 5.

4 EMPIRICAL STATUS AND FALSIFICATION PATH

No current empirical claim is made. PLAN-ENABLE-001 specifies a future falsification experiment for a training-visible enable policy in two real scenarios: natural human-label noise and a clean benign-hard cohort. It must freeze code, data and split hashes, comparator arms, seeds, environment, event schema, and forbidden-field checks before execution. Success on one scenario alone would not establish a deployment claim; failure is an informative boundary for that particular policy, not a refutation of T1–T3.

5 RELATED WORK

Influence functions (Koh & Liang, 2017), datamodels (Ilyas et al., 2022), and TRAK (Park et al., 2023) provide relevant ways to relate training data to learned-model behavior. SISA (Bourtoule et al., 2021) is a different retraining-based unlearning construction. Work on SGD noise and implicit regularization (Mandt et al., 2017; Smith & Le, 2018; Smith et al., 2021) motivates stating the conditional-mean premise in T3 rather than treating it as an SGD identity. Loss-mixture and noisy-label work (Arazo et al., 2019; Jiang et al., 2018; Han et al., 2018; Wei et al., 2022) motivates the planned experiment, not a present policy result. The closest-work matrix records the checked metadata and the deliberately narrow relationship claims.

6 LIMITATIONS

The scope is mathematical rather than empirical. T1 is fixed quadratic. T2 requires smooth strong convexity along the relevant segments, a stable step-size bound, and a separate invariant-subspace condition for its directional refinement. T3 requires the affine scalar recursion and conditional-zero-mean innovation. None of these assumptions automatically holds for logistic training, shuffled SGD, neural networks, noisy labels, or data-selection systems. The public package also lacks a same-version experiment evidence chain; remote candidate results are not attributed to this manuscript.

7 CONCLUSION

The three results establish exact or bounded product relations only within their stated regimes. Their shared form motivates careful future testing of path interventions, but it does not license a broader theory or a deployment recommendation. The next empirical step is a registered, matched-path falsification test, not recovery of unsupported historical numbers.

REPRODUCIBILITY STATEMENT

This self-contained source package includes its TeX dependencies and bibliography and is built in isolation. Formal assumptions, statements, and proofs are active in this source. The separate execution-readiness audit names the artifacts required before PLAN-ENABLE-001 can produce evidence; no unexecuted protocol or remote-only result is reported as an observation.

ETHICS STATEMENT

This revision reports no newly run human-subject or dataset study. Any future execution must verify dataset licenses, versions, and use conditions, and the policy must not access clean, protected, test, or oracle labels at decision time.

AI USE STATEMENT

This manuscript and its audit package were prepared with substantial assistance from LLM-based coding agents, including Codex and earlier multi-agent research workflows, under human supervision. The use of such systems included drafting, revision, repository inspection, provenance and citation checks, build checks, and internal review planning. No LLM-generated assertion was treated as evidence without an identified source artifact or an explicit “planned” label.

Tasks performed with AI assistance.

- Drafting and revising prose, equations, figures, tables, and this disclosure; generating alternative paper stories and revision plans.
- Inspecting the public baseline and a separately frozen remote candidate; computing hashes and diffs; checking source-package dependencies; and running safe, read-only consistency/configuration verifiers.
- Searching for and cross-checking primary or official bibliographic metadata, then building a claim–evidence map and closest-work matrix.
- Designing (but not submitting or executing) the planned matched-path experiment and auditing its data, environment, launcher, comparator, seed, and output-schema readiness.
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- Human supervisors specify the task, authorization boundaries, resource limits, and submission decisions. This revision did not authorize a GPU, remote, or formal experimental run.
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A PROOF-SCOPE CHECKLIST

T1 uses a fixed symmetric positive-definite Hessian and its eigendecomposition. T2 uses segment Hessians bounded between μI and LI ; its directional line is conditional on simultaneous invariance. T3 takes conditional expectations of the stated scalar recursion. These are the only proof dependencies used by the active manuscript. Historical figures, tables, scripts, and numerical claims are not imported, hidden, or compiled here.